

MEMORY DIVERGENCE IS NOT BEHAVIORAL DIVERGENCE: STAGE-RESOLVED TRANSPORT IN SELF-IMPROVING AGENT MEMORY

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Paper under double-blind review

ABSTRACT

Self-improving agents increasingly convert terminal feedback into persistent memory, yet common evaluations conflate whether feedback changes stored state, whether that state is retrieved, whether the policy uses the branch-specific difference, and whether behavior changes. We freeze each source trajectory, switch only the released success/failure writer branch, and measure write W , native source-item exposure E , first-action uptake U , terminal outcome O , plus a forced-capacity control F . Writing diverges for all 20 Shopping and 4 Reddit pairs; forced supply has downstream leverage ($|\Delta| = 0.15625$, $p = 0.00074$). Under native Shopping reuse, however, source memory is retrieved in 125/172 opportunities while first-action uptake is unsupported (TV = 0.06944, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). The prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ with independent F induces a 10-state diagnostic class; exhaustive enumeration of all 2^5 observation subsets shows that only the full $W/E/U/O/F$ basis separates every state. We then test repair actionability prospectively. A matched explicit memory-use reminder has no aggregate effect on 13 outcome-blind states (312/312 calls; $U_{A0} = U_{A1} = U_{A2} = 0.0962$). On a separate fresh 12-state pilot, an intentionally non-novel state-conditioned binding adapter raises mean uptake from 0.125 natively and 0.083 under a memory-only adapter to 0.208; however, the gain is heterogeneous (mean $U_{A2} - U_{A1} = 0.125$, positive in only 3/12 states), so its preregistered consistency gate fails. Thus stage resolution can identify a repair family worth probing and produce a conditional transport opening, but it does not by itself supply a universal repair prescription. Persistent-state divergence is not behavioral authority, availability is not use, and repair authority requires separate intervention evidence.

1 INTRODUCTION

A persistent-memory agent can change internally long before task accuracy reveals anything. Consider two copies of the same interaction trajectory. If one is labeled successful and the other failed before reflection, an outcome-conditioned writer can store different reusable advice although the underlying experience is fixed. That establishes a persistent-state intervention, but not behavioral self-improvement: the different state must still be exposed by retrieval, influence a decision, and survive to an outcome.

We study this gap as *stage-resolved transport*. For a frozen trajectory τ , we switch the released success/failure reflection branch—a bundled writer-protocol intervention that changes reflection instruction together with outcome/reward semantics—and instrument

write (W) $\rightarrow$ source-item exposure (E) $\rightarrow$ first-action uptake (U) $\rightarrow$ outcome (O).

A separate forced fixed-evidence arm F bypasses native retrieval and asks whether the paired memory content can affect the downstream system when supplied. These surfaces answer different questions and have different units. A write difference proves state change, a retrieval hit proves availability of the source item, and neither proves that the branch-differentiating residual changed the policy. Conversely, a weak endpoint alone cannot tell whether the update failed to write, failed to retrieve,

was ignored after retrieval, or changed behavior only sparsely. We call this loss of mechanism information *evaluation coarsening*.

The frozen evidence produces exactly this disagreement across surfaces. Reward-conditioned writing diverges for 20/20 Shopping pairs and 4/4 Reddit pairs; a stronger same-mode wording control leaves a reward-mode excess of 0.105 ($p = 0.0078$). Forced supply yields terminal $|\Delta| = 0.15625$ ($p = 0.00074$). Yet native Shopping retrieves the branch-conditioned source item in 125/172 held-out opportunities while the first-action branch contrast is not supported (TV = 0.06944, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). Reddit again has reliable write divergence but 6/8 zero terminal cells and opposite signs in the two nonzero cells. The strongest supported operational localization is therefore after source-item exposure and before stable first-action uptake. It is an evidence boundary, not a causal attenuation onset or mediation coefficient.

Stage resolution is not merely “report more metrics.” The native prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ permits five binary prefix depths; crossing them with $F \in \{0, 1\}$ yields a complete 10-state diagnostic class for this experiment. Exhaustive enumeration of all 32 observed-surface subsets shows that $W/E/U/O/F$ is the unique separating basis: omitting a native coordinate aliases adjacent prefix states, while omitting F aliases the two capacity states at every depth. This paper-specific result explains why coarsening can make genuinely different failure explanations observationally identical.

The diagnostic also creates an engineering question: does the first unsupported measured stage identify a useful repair family? We test this prospectively in two steps rather than assuming a repair. First, on 13 outcome-blind states we compare native prompting with matched decision-check controls; an explicit reminder to use memory leaves mean uptake unchanged ($U_{A0} = U_{A1} = U_{A2} = 0.0962$; 312/312 calls complete), so salience alone does not repair the boundary. Second, on a disjoint fresh 12-state pilot we compare native memory, a memory-only adapter, and a state-conditioned binding adapter that converts the same retrieved memory into a concise current-state action implication before policy execution. The adapter is deliberately a proof-of-concept, not a novel memory architecture. Mean uptake rises to 0.208 from 0.125 natively and 0.083 under memory-only adaptation, with mean $D = U_{A2} - U_{A1} = 0.125$. The effect is nevertheless concentrated: $D > 0$ in only 3/12 states, 4 are negative, and the preregistered cross-state consistency gate fails. The combined result is sharper than either pilot alone: *the diagnosis selects a mechanism-aligned repair family and state binding can reopen transport in some states, but repair authority remains conditional rather than automatic*.

Our contributions are:

1. **Controlled stage-resolved transport.** A same-trajectory success/failure writer intervention follows one induced persistent-state contrast through native exposure, first-action uptake, outcome, and an explicit forced-capacity bypass control.
2. **Diagnostic completeness under the experiment’s prerequisite class.** The complete 10-state class induced by $O \Rightarrow U \Rightarrow E \Rightarrow W$ and F requires the full $W/E/U/O/F$ observation basis to avoid aliases.
3. **An empirical transport boundary.** Write divergence and source-item exposure are established while stable first-action uptake is not, and cross-domain terminal effects are sparse and sign-heterogeneous.
4. **Two prospective repair probes that separate salience from state-conditioned binding.** A simple explicit-use reminder has no aggregate effect, while a fresh state-conditioned adapter increases mean first-action uptake relative to native and memory-only controls but fails the preregistered cross-state consistency gate. This establishes a heterogeneous proof-of-concept opening without promoting the adapter to a universal repair or a new method claim.

The resulting claim is deliberately narrower than a new memory-utilization method: **state divergence is not behavioral authority; availability is not use; diagnosis is not a repair prescription.**

2 STAGE-RESOLVED MEMORY TRANSPORT

2.1 CONTROLLED WRITE INTERVENTION AND NATIVE OBSERVABLES

Let τ be a frozen interaction trajectory before memory construction. We create two writer conditions

$$r \in \{S, F\}, \quad m^{(r)} = g(\tau, r), \quad W(\tau) = d_{\text{mem}}(m^{(S)}, m^{(F)}),$$

where S/F selects the released success- or failure-reflection branch. $W > 0$ establishes an operational persistent-state contrast, not atom-level causal purity of every generated sentence; decoding-seed/noise-floor replication is unavailable, and the branch jointly changes reflection instruction and outcome/reward semantics. Section 4 therefore tests a stronger generic-wording reduction.

For a matched held-out future-task state q , native source-item exposure is

$$E_r(q) = \mathbf{1}[m^{(r)} \in \text{Retrieve}(q; \mathcal{M}^{(r)})].$$

This proves that the branch-conditioned source item reached policy context. It is weaker than *treatment-residual exposure*: the frozen logs do not show that the branch-differentiating content itself was represented or consumed downstream. Let $p_r^{(1)}(\cdot | q)$ be the empirical distribution over normalized first structured actions for the same matched future-task state; its support is the union of actions observed in the two branches. We measure

$$U(q) = \text{TV}(p_S^{(1)}(\cdot | q), p_F^{(1)}(\cdot | q)), \quad O(q) = |y_S(q) - y_F(q)|.$$

A *branch-comparison state* is one matched future-task state evaluated under paired success/failure memories while non-memory task state is fixed; the 36 Shopping states are 36 such units, not 36 independent source trajectories. The native object is therefore (W, E, U, O) , whose components are ordered but not commensurate.

2.2 FORCED CAPACITY IS A SIDE CONTROL

The forced fixed-evidence arm mechanically supplies branch-specific memory and measures

$$L_{\text{forced}}(q) = |y^{\text{forced}, S}(q) - y^{\text{forced}, F}(q)|.$$

It bypasses native retrieval and therefore is not inserted between W and E . A positive F weakens global downstream-insensitivity explanations; it cannot make an unsupported native stage pass.

2.3 DIAGNOSTIC COMPLETENESS FROM THE PREREQUISITE CLASS

Let $\mathcal{B} = \{W, E, U, O, F\}$ with native prerequisite order

$$O \Rightarrow U \Rightarrow E \Rightarrow W.$$

A prerequisite-consistent binary diagnostic state has native prefix depth $p \in \{0, 1, 2, 3, 4\}$ and forced-capacity bit $f \in \{0, 1\}$:

$$v(p, f) = (\mathbf{1}[p \geq 1], \mathbf{1}[p \geq 2], \mathbf{1}[p \geq 3], \mathbf{1}[p \geq 4], f).$$

Thus the experiment induces exactly 10 idealized states. For observed surfaces $A \subseteq \mathcal{B}$, let Π_A retain only coordinates in A .

Prerequisite-class diagnostic completeness. Π_A is injective over all 10 states iff $A = \mathcal{B}$. The full basis is injective. If any native coordinate is omitted, its two adjacent prefix depths alias separately for $F = 0$ and $F = 1$; if F is omitted, $(p, 0)$ and $(p, 1)$ alias for all five depths. Exhaustive enumeration of all $2^5 = 32$ surface subsets verifies that $W/E/U/O/F$ is the unique separating basis for this class (Appendix D). This is a paper-specific identifiability result, not a universal memory taxonomy or generic separating-set novelty, and empirical NOT-SUPPORTED evidence is never coerced into literal zero.

2.4 ORDINAL STAGE EVIDENCE AND INTERPRETATION

Because W , exposure rate, first-action TV, and terminal $|\Delta|$ have different units, samples, and test sensitivities, we do not construct a scalar transport coefficient. Let native stages be $s_1 = W, s_2 = E, s_3 = U, s_4 = O$. Each retains its own state—SUPPORTED, directly OBSERVED, or NOT-SUPPORTED—and we report

$$b^* = \min\{j : s_j \text{ is not supported/observed while } s_1, \dots, s_{j-1} \text{ are supported/observed}\}.$$

This operator localizes the first unsupported *measured* stage; it does not estimate where a latent causal effect physically attenuates. Stage-local tests are not searched for the smallest p -value and do not form an omnibus “at least one stage is significant” claim, so multiplicity is not used post hoc to select b^* .

The current evidence gives supported write, observed source-item exposure, unsupported first-action uptake, and sparse/sign-heterogeneous outcomes, hence $b^* = U$. Three distinctions are load-bearing:

state divergence $\nRightarrow$ behavioral divergence,
source-item exposure $\nRightarrow$ policy uptake,
forced capacity $\nRightarrow$ native transport.

They support a stage-indexed credit rule: W can justify state-change credit, E availability credit, a supported U behavioral-control credit, and outcome evidence utility validation. Crucially, this rule governs *what evidence may justify*; it does not imply that the first unsupported stage already identifies a successful repair. Section 5.4 tests that stronger engineering prediction prospectively.

3 STAGE 1: REWARD-CONDITIONED WRITING ROBUSTLY CHANGES PERSISTENT STATE

We first test the write stage with paired interventions over released WebArena Shopping trajectories. Within each pair, the task text and full action history are fixed before the writer is called. One branch uses the released success-reflection objective and the other uses the released failure-reflection objective; the resolved writer model is matched within pair and temperature is zero. Raw responses and the pre-writer trajectory projection are content-addressed before analysis.

The initial F0 tranche requested six source trajectories. Four returned complete memory outputs in both conditions; the two incomplete requests terminated in the failure-label writer before assistant text existed and were excluded from the paired content estimand. Among the four complete pairs, every pair changed normalized memory content and the extracted memory-item title set, with mean token-set Jaccard distance 0.735. Receipt-level forensics preserve the two incomplete arms as provider-response failures rather than treating them as negative memory examples.

The later frozen breadth tranche expands the write-stage evidence to 20 complete Shopping success/failure pairs. All 20/20 pairs produce different branch memories, with pooled token-set Jaccard distance 0.673. When combined with the four Reddit pairs used in the cross-domain replication, the D0 lineage contains 24/24 paired source experiences with the same pre-writer trajectory projection, the same resolved writer model within pair, temperature zero, and different branch-memory content. This establishes a robust operational branch contrast at the persistent-state level.

The content changes are not only token-level. A deterministic post-ready strategy-slot audit maps frozen memory titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. In the original four complete F0 pairs, the strategy-slot set changes in all four. We treat this as a structural diagnostic rather than a semantic-validity claim: the current data do not certify each textual delta atom as a pure reward-caused proposition because explicit decoding seeds or same-condition noise-floor replications are unavailable.

The write-stage result therefore supports a narrow statement: under the released writer policy, flipping the reward-conditioned reflection branch on the same source experience reliably produces a different durable state. It does not by itself establish that the difference will be retrieved, followed, or reflected in terminal behavior. Section 4 first tests the strongest simple prompt-sensitivity reduction; Section 5 then measures downstream transport stage by stage.

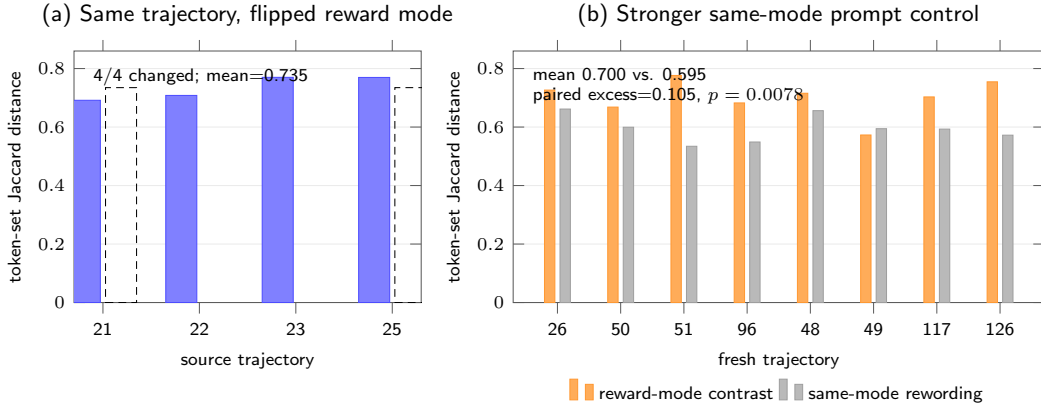


Figure 1: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

4 RULING OUT GENERIC PROMPT SENSITIVITY

The success/failure writer-branch intervention changes the reflection instruction together with the outcome/reward semantics. It is therefore a bundled protocol contrast, not a pure reward-bit intervention. A generic prompt-sensitivity account predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence; the following control tests that reduction, but cannot decompose instruction semantics from reward semantics within the released branch. We evaluate it on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewodings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8} \sum_{i=1}^8 (B_i - W_i). \quad (1)$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

5 STAGE-RESOLVED TRANSPORT EVIDENCE

We combine forced supply, native source-item exposure, first-action uptake, and terminal outcomes because each rules on a different failure explanation.

Table 1: Stage evidence. The forced arm is a side capacity control, not a native stage.

Stage	Evidence state	Frozen observation / interpretation
Write	Supported	20/20 Shopping pairs diverge; same-mode excess 0.105, $p = 0.0078$.
Forced capacity	Supported	$ \Delta = 0.15625$, $p = 0.00074$; downstream leverage under supplied memory.
Source-item exposure	Observed	125/172 retrieval hits; availability, not residual consumption or use.
First-action uptake	Not supported	TV = 0.06944, $p = 0.5801$, 0/36 modal changes.
Native outcome	Sparse / heterogeneous	Shopping 34/36 zero; Reddit 6/8 zero with opposite signs in the two nonzero cells.

5.1 FORCED SUPPLY ESTABLISHES DOWNSTREAM CAPACITY

We first bypass native retrieval and directly inject paired memories into fixed future-task packets. Four complete initial source-memory pairs are crossed with four frozen future tasks, with eight fresh rollouts per memory condition in each of 16 cells (256 completed calls). Mean absolute terminal success-rate difference is 0.15625, above the frozen 0.15 practical floor, with a 100,000-draw within-cell permutation $p = 0.00074$. Thus the downstream system can respond to supplied branch-specific memory. This is capacity, not native transport: eight cells are zero and the signed effects are heterogeneous, so even forced supply does not support a universal harmful direction.

5.2 NATIVE EXPOSURE IS COMMON WHILE FIRST-ACTION UPTAKE IS UNSUPPORTED

Restoring the native memory-bank path, the branch-conditioned source item is retrieved in 125/172 held-out Shopping opportunities (0.727). This is *source-item exposure*, not treatment-residual exposure: retrieval proves the carrier item entered context, not that its branch-differentiating content was consumed. The native null therefore cannot be summarized as “the source memory was never retrieved.”

Across 36 matched Shopping branch-comparison states, mean first-action TV is 0.06944 with $p = 0.5801$, and 0/36 states change their modal first structured action. This is deliberately an early decision probe; it does not rule out later tool-sequence or plan-level differences. Native terminal transport is also small: mean absolute success difference is 0.02083 ($p = 0.4289$), with 34/36 cells exactly zero. Hence first-action uptake is the first unsupported *measured* native stage after supported write and observed source-item exposure. This is an evidence-localization boundary, not a causal onset or mediation coefficient. Removing terminal evidence, Reddit, or the forced side control leaves this boundary unchanged; merging exposure with uptake destroys the distinction.

5.3 CROSS-DOMAIN AND ENDPOINT BOUNDARIES

On Reddit, reward-conditioned writing diverges in 4/4 source pairs, but native terminal $|\Delta| = 0.125$ remains unresolved ($p = 0.2253$): 6/8 cells are zero and the two nonzero effects have opposite signs. Reddit therefore supports the separation between write divergence and stable native outcome transport, not the Shopping first-action boundary because matched exposure/action measurements are unavailable. An outcome-blind structured-memory control likewise yields a statistically detectable but small terminal contrast (0.04514, $p = 0.0048$), below the frozen 0.15 practical floor with 32/36 zero cells. Statistical detectability alone is not behavioral authority.

5.4 PROSPECTIVE REPAIR TEST: DIAGNOSIS IS NOT A PRESCRIPTION

The stage signature $W^+ / E^{\text{obs}} / U^{\text{ns}} / F^+$ suggests that post-retrieval policy integration is the next surface to *probe*; it does not imply a successful repair. We prospectively freeze 13 pilot states—one outcome-blind selection per intent template from the 36 replayable states—and seal the remaining 23 states as an untouched confirmatory holdout. Upstream task state, success/failure memories, retrieved packet bytes/order, model identity, decoding, and first-action parser are fixed. We compare native prompting (A0), a memory-blind extra decision check (A1), and a syntax-matched explicit check

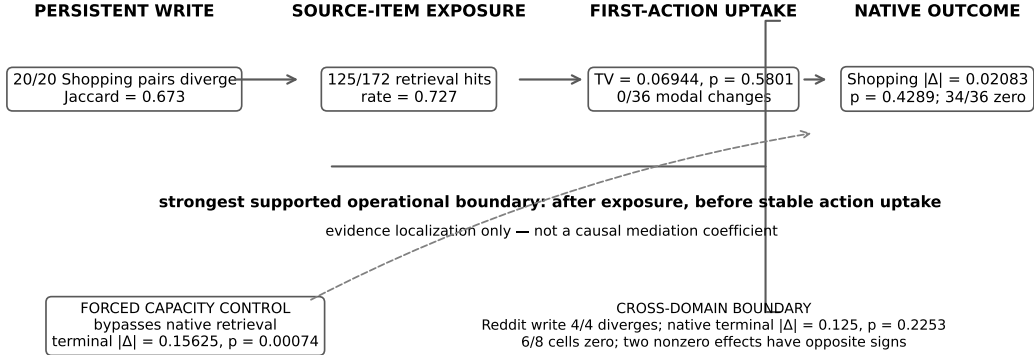


Figure 2: **Native transport and forced capacity are different surfaces.** Write and source-item exposure are established, while first-action uptake is the first unsupported measured native stage. Forced capacity remains a bypass control.

asking whether reusable memory should change the next action (A_2). Each state receives four rollouts per branch and arm, for 312 calls; all 312 complete with no model, packet, or prompt-hash drift.

The preregistered pilot requires mean $U_{A_2} \geq 0.20$, mean $D = U_{A_2} - U_{A_1} \geq 0.10$, $D > 0$ in at least 8/13 states, and mean $N = U_{A_2} - U_{A_0} > 0$. It fails without threshold changes: mean $U_{A_0} = U_{A_1} = U_{A_2} = 0.09615$, mean $D = 0$, mean $N = 0$, and D is positive in 3/13 states, negative in 2/13, and zero in 8/13 (descriptive bootstrap 95% CI for mean D : $[-0.115, 0.096]$). The explicit memory-use check occasionally perturbs actions, but not in a systematic branch-specific direction. Per the frozen rule, the 23-state confirmatory experiment is not run.

This negative result separates three claims that are easy to conflate. The stage-resolved observations and diagnostic-completeness result remain intact; the stage-indexed credit rule still says that write/exposure cannot earn behavioral credit; but the stronger hypothesis that localization plus an explicit-use reminder is itself a repair is not supported. **Diagnosis identifies where evidence weakens; salience alone does not repair that stage.** Appendix E reports the pilot design and per-state outcomes.

5.5 MECHANISM-ALIGNED PROOF OF CONCEPT: STATE-CONDITIONED BINDING

We next ask a narrower question on a disjoint fresh state set: whether adapting retrieved memory to the *current state*, rather than merely reminding the policy that memory exists, can reopen first-action transport. We intentionally do not claim this adapter as method novelty; state-aware memory and target-conditioned reuse are established design families. From 86 retrieval-hit Shopping states with available released trajectories that exclude all previous 36 branch-comparison states, we select one state per fresh intent template by a frozen hash rule, then freeze 12 pilot states and 19 untouched template holdouts. The policy experiment never reads terminal outcomes.

For each branch memory we compare: A_0 , the native retrieved memory; A_1 , a memory-only adapter that rewrites the same memory into a concise procedural note without target task/state; and A_2 , a state-conditioned binder that sees the same memory, target task, and current browser state and emits one concise current-state action implication. The binder is frozen before policy rollouts and uses the same resolved model at temperature zero; the acting policy remains unchanged at temperature 0.2. Each state receives six first-action rollouts per branch and arm. All 48 binder calls and all 432 policy calls complete with no model drift and no use of the 19-state holdout.

Mean branch-specific uptake is $U_{A_0} = 0.125$, $U_{A_1} = 0.0833$, and $U_{A_2} = 0.2083$. Thus the state-conditioned arm shows an aggregate increase of 0.125 over the memory-only adapter and

0.0833 over native reuse. This is not a general repair pass: the preregistered pilot also required $D_i = U_{i,A2} - U_{i,A1} > 0$ in at least 6/12 states, whereas only 3/12 are positive, 4/12 are negative, and 5/12 are zero. A descriptive state bootstrap for mean D gives 95% interval $[-0.083, 0.375]$, and the three largest squared D_i values contribute 90.9% of squared effect mass. One binder output also violates the requested 60-word format, although that state has zero uptake in all three arms and does not drive the positive cells.

The engineering conclusion is therefore deliberately modest: **state-conditioned binding can open the diagnosed exposure-to-uptake transport path in a subset of fresh states, but the effect is heterogeneous and does not earn universal repair authority.** The contrast with the failed explicit-use reminder matters more than the mean gain itself: simply increasing memory salience does not move uptake, while a mechanism-aligned state-binding transformation can. We leave the fresh 19-template holdout untouched rather than tuning an applicability gate after observing this pilot.

6 RELATED WORK

Self-evolving memory and management. ReasoningBank distills reusable reasoning from self-judged successful and failed experience (Ouyang et al., 2026); reward-conditioned construction and reward/reuse distortion are studied elsewhere (Wang et al., 2025; Asadolahi et al., 2026; Yang et al., 2026). Xiong et al. (2026) analyze experience-following, error propagation, and quality-aware addition/deletion. C1 does not claim feedback-driven memory or generic governance as new; it creates two persistent states from the *same* source trajectory and asks how far that controlled contrast survives.

Retrieval, utilization, and state-conditioned memory. Mem2ActBench directly evaluates active memory application (Shen et al., 2026); MemCoRL co-optimizes retrieval and utilization (Liu et al., 2026); Chain-of-Memory strengthens post-retrieval utilization (Xu et al., 2026); and SAMem explicitly aligns fine-grained memory with the agent’s current state (Wang et al., 2026). QCR and other diagnostics likewise separate retrieval from later reuse (Li et al., 2026; Yuan et al., 2026). Thus “retrieval is not use,” state-aware memory, and utilization optimization are not our novelty. Our residual is controlled transport identification: preserve the intervention identity across $W/E/U/O/F$, determine what stage-specific evidence can justify, and use known-style post-retrieval transformations only as prospective probes of whether the diagnosed surface is engineering-relevant. The repair probes therefore validate or bound the diagnosis rather than introduce a new utilization architecture.

7 LIMITATIONS

The success/failure writer branch is a bundled protocol contrast, not a pure reward-bit intervention; decoding-seed/noise-floor replication is unavailable and two initial failure-label calls produced no text. Native exposure is source-item availability rather than identified treatment-residual consumption. Stage tests differ in units and power, so the first-action boundary is operational evidence localization, not causal mediation or proof that later policy surfaces cannot differ. Shopping and Reddit remain domain-limited, and other writers/backbones are untested.

The repair evidence has two different boundaries. Prompt hashes establish exact treatment *delivery*, not internal cognitive compliance, and the simple explicit-use reminder is unsupported as an actionable repair. The later state-conditioned pilot uses 12 disjoint fresh states and shows a higher aggregate uptake mean, but its preregistered cross-state consistency gate fails: only 3/12 states improve over the memory-only adapter, four worsen, and three cells dominate 90.9% of squared contrast mass. We therefore treat state binding as a heterogeneous proof of concept rather than a validated general repair. One state-conditioned binder output also violates the requested length format, although that state has zero uptake in all arms. Neither the old 23-state holdout nor the new 19-template holdout is consumed after these pilot results.

8 CONCLUSION

The same persistent update looks strong at write, active at source-item retrieval, weak at measured native action/outcome, and consequential under forced supply. The prerequisite class explains why observing fewer than *W/E/U/O/F* aliases distinct diagnoses. Engineering probes then refine what the diagnosis means: a simple memory-use reminder does nothing, whereas a known-style state-conditioned binding adapter raises average first-action uptake on fresh states but only heterogeneously and fails the preregistered consistency gate. The practical rule is therefore two-stage: **credit only what the strongest supported stage justifies, then demand separate intervention evidence before granting repair authority**. State divergence is not behavioral authority; availability is not use; stage diagnosis can point to a repair family without making that repair universal.

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A F0 REPRODUCTION DETAILS

A.1 RELEASED TRAJECTORY SAMPLE

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 MEMORY CONSTRUCTION INTERVENTION

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

A.3 PAIR-LEVEL F0 RESULTS

Table 2: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 3 reports the paired distances used in the frozen sign-flip test.

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

Table 3: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C.2 FROZEN TERMINAL MATRIX

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source-future cells. Each cell receives eight fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

Table 4: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$ \Delta $
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
Mean absolute difference				0.15625

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the revised paper treats the experiment as a forced-leverage measurement rather than a uniform directional effect or a native end-to-end transport estimate.

If reward-label corruption selects the failure-conditioned memory with probability q , the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c . The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$. This remains a descriptive plug-in calculation conditional on forced memory exposure; it is not used as the main claim in the revised stage-resolved paper.

C.4 NATIVE TRANSPORT AUDIT ADDED AFTER THE FORCED-LEVERAGE STUDY

The later frozen native-pipeline audit changes the interpretation of the forced matrix. In Shopping, the branch-conditioned source-memory item is retrieved in 125/172 held-out opportunities after bank expansion, but the mean absolute native terminal success difference is only 0.02083 with permutation $p = 0.4289$; 34/36 terminal cells are zero. The corresponding first-action total-variation statistic is 0.06944 with $p = 0.5801$, and no frozen state changes its modal first action between reward branches. A post-hoc working-memory branch-shift diagnostic is 0.00335 with $p = 0.2052$.

The cross-domain Reddit audit contains four complete success/failure writes, all of which diverge. Its native terminal mean absolute difference is 0.125 with $p = 0.2253$; 6/8 cells are zero and the two nonzero cells have opposite signs. These later measurements are why the main text separates forced capacity from native transport and treats first-action uptake as the first unsupported measured Shopping stage after write and source-item exposure, rather than as an identified latent attenuation onset or causal mediation result.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source-future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.5 FULL ALTERNATIVE-EXPLANATION AUDIT

Table 5: Alternative-explanation audit. “Weakened” means the frozen evidence makes the explanation incomplete but does not mathematically eliminate every variant of it.

Explanation	Frozen evidence	Disposition
No persistent state intervention	20/20 Shopping and 4/4 Reddit writes diverge; same-mode excess = 0.105, $p = 0.0078$	Inconsistent with the observed write-stage intervention.
Downstream policy is globally insensitive to memory	Forced terminal $ \Delta = 0.15625$, $p = 0.00074$	Weakened: supplied branch-specific memory has measurable leverage.
Native null is only retrieval absence	125/172 native Shopping source-item retrieval hits	Weakened: substantial item exposure coexists with weak first-action contrast.
Retrieval hit is a surrogate for policy use	first-action TV = 0.06944, $p = 0.5801$; 0/36 modal changes	Rejected as an evaluation equivalence: item availability is not policy uptake.
Universal directional terminal transport	Shopping 34/36 zero; Reddit 6/8 zero, two nonzero signs opposite	Not supported by the frozen native cells.
Causal mediation has been identified	no per-atom seed binding/noise-floor replication; forced/native surfaces differ	Unresolved and explicitly outside the claim set.

D PREREQUISITE DIAGNOSTIC STATE CLASS

The native prerequisite order $O \Rightarrow U \Rightarrow E \Rightarrow W$ permits five binary prefix depths. Crossing each with forced-capacity bit $F \in \{0, 1\}$ yields the 10-state class used in Section 2.3.

A deterministic content-addressed verifier enumerates all 32 subsets of $\{W, E, U, O, F\}$. Only the full set is injective over all 10 states. Omitting W, E, U , or O aliases two adjacent-prefix pairs (one at each fixed F); omitting F aliases five pairs. This is exact for the stated binary prerequisite class only. It does not convert empirical NOT-SUPPORTED evidence into zero or claim universal diagnostic optimality for arbitrary stochastic memory systems.

Table 6: Native prefix states; each row occurs with both $F = 0$ and $F = 1$.

Depth	Interpretation	W	E	U	O
0	no native branch-specific state	0	0	0	0
1	write only	1	0	0	0
2	written and source-item exposed	1	1	0	0
3	first-action uptake	1	1	1	0
4	terminal transport	1	1	1	1

E PROSPECTIVE REPAIR PILOT DETAILS

The pilot is selected outcome-blind from the 36 replayable Shopping states: states are partitioned by intent-template ID and exactly one state per template is chosen by the lexicographically smallest SHA256 of a frozen salt, template ID, and task ID. The 13 selected states and the remaining 23-state confirmatory holdout are disjoint; the holdout receives no new $A1/A2$ calls after pilot outcomes are observed.

All arms reuse byte-identical upstream branch memories and task-state packets. $A0$ uses native prompting. $A1$ adds a memory-blind check asking whether the ultimate task and current browser state change the next action. $A2$ adds a syntax-matched check asking whether reusable memory and current browser state change the next action. Each arm receives four rollouts for each success/failure memory branch per state. The primary memory-specific contrast is $D_i = U_{i,A2} - U_{i,A1}$; $N_i = U_{i,A2} - U_{i,A0}$ is secondary. The gate is frozen before execution: mean $U_{A2} \geq 0.20$, mean $D \geq 0.10$, $D_i > 0$ in at least 8/13 states, and mean $N > 0$.

Table 7: Full 13-state repair pilot. $D = A2 - A1$ and $N = A2 - A0$ are first-action TV contrasts.

Task	Template	Source	U_{A0}	U_{A1}	U_{A2}	D / N
164	136	163	.50	.25	.50	+.25 / .00
329	147	141	.00	.00	.00	.00 / .00
150	155	226	.25	.00	.25	+.25 / .00
124	159	226	.00	.00	.00	.00 / .00
321	160	188	.00	.00	.00	.00 / .00
142	162	141	.00	.50	.00	-.50 / .00
279	204	226	.50	.25	.25	.00 / -.25
360	206	231	.00	.00	.00	.00 / .00
233	213	232	.00	.00	.00	.00 / .00
190	214	189	.00	.00	.00	.00 / .00
26	222	25	.00	.00	.00	.00 / .00
230	370	226	.00	.25	.00	-.25 / .00
384	666	25	.00	.00	.25	+.25 / +.25
Mean			.096	.096	.096	.000 / .000

All 312 requests complete, with zero model-identity or prompt-hash drift and no arm-dependent missingness; four responses require the frozen parser’s regex fallback. D_i is positive in 3 states, negative in 2, and zero in 8; the descriptive bootstrap 95% interval for mean D is $[-0.115, 0.096]$. The pilot therefore stops the current treatment before confirmatory scaling. Exact prompt delivery verifies treatment realization at the input surface, not internal cognitive compliance; the supported conclusion is only that this simple explicit memory-use-check realization does not selectively increase branch-specific uptake.

F FRESH STATE-CONDITIONED BINDING PILOT

To avoid reusing either the 13-state prompt pilot or its sealed 23-state holdout, the state-conditioned pilot starts from 86 additional Shopping states that have released trajectories and a retrieval hit under the frozen expanded bank. These 86 states span 31 fresh intent templates. We select one state per template by a frozen hash rule and then choose 12 pilot states by a second frozen hash; the remaining 19 template representatives receive no binder or policy calls after pilot outcomes are observed. The 12 pilot states cover nine selected source memories.

The binder uses the frozen Doubao Seed 2.0 Mini model at temperature zero. $A1$ receives only the retrieved memory and rewrites it into one concise procedural note. $A2$ receives the same memory together with the target task and current browser state and emits one concise current-state action implication. The policy model, retrieved raw memory, state packet, decoding, and action parser are otherwise unchanged. The policy receives six rollouts per success/failure branch and arm, for 432 action calls after 48 binder calls. All calls complete without model drift; the prior 36 states and the 19 fresh holdouts receive zero calls.

Table 8: Fresh 12-state state-conditioned binding pilot. $A1$ is memory-only adaptation; $A2$ is state-conditioned binding.

Task	Template	Source	U_{A0}	U_{A1}	U_{A2}	D / N
794	191	188	.000	.000	.000	.000 / .000
161	171	159	.667	.167	1.000	+.833 / +.333
336	169	232	.000	.000	.000	.000 / .000
531	154	188	.000	.167	.000	-.167 / .000
276	212	158	.333	.000	.500	+.500 / +.167
586	194	141	.167	.000	1.000	+1.000 / +.833
235	213	232	.000	.000	.000	.000 / .000
286	207	158	.167	.167	.000	-.167 / -.167
432	145	226	.167	.333	.000	-.333 / -.167
359	206	231	.000	.000	.000	.000 / .000
47	197	189	.000	.000	.000	.000 / .000
326	208	25	.000	.167	.000	-.167 / .000
Mean			.125	.083	.208	+.125 / +.083

The preregistered pilot signal requires mean $D \geq 0.05$, mean $N > 0$, U_{A2} above both controls, and $D_i > 0$ in at least 6/12 states. The first three conditions pass, but only 3/12 states have positive D_i (4 negative, 5 zero), so the overall gate fails. A descriptive state bootstrap gives mean- D 95% interval $[-0.083, 0.375]$, and the three largest squared D_i values account for 90.9% of squared effect mass. A post-hoc lexical divergence between success- and failure-branch $A2$ notes correlates with D_i at $r = 0.725$; this is hypothesis generation only and is not used to reopen the untouched holdout.